

SCHOOL OF ENGINEERING

Diploma in Electronics & Computer Engineering (EGDF20)

Wireless Communication & Networking (EGE321)

Tutorial 5: Link Performance Analysis

- 1) What are the 2 factors that determine the path loss in Free Space propagation model?

Distance and frequency

- 2) In practical scenarios, name other factors that also have an impact on path loss.

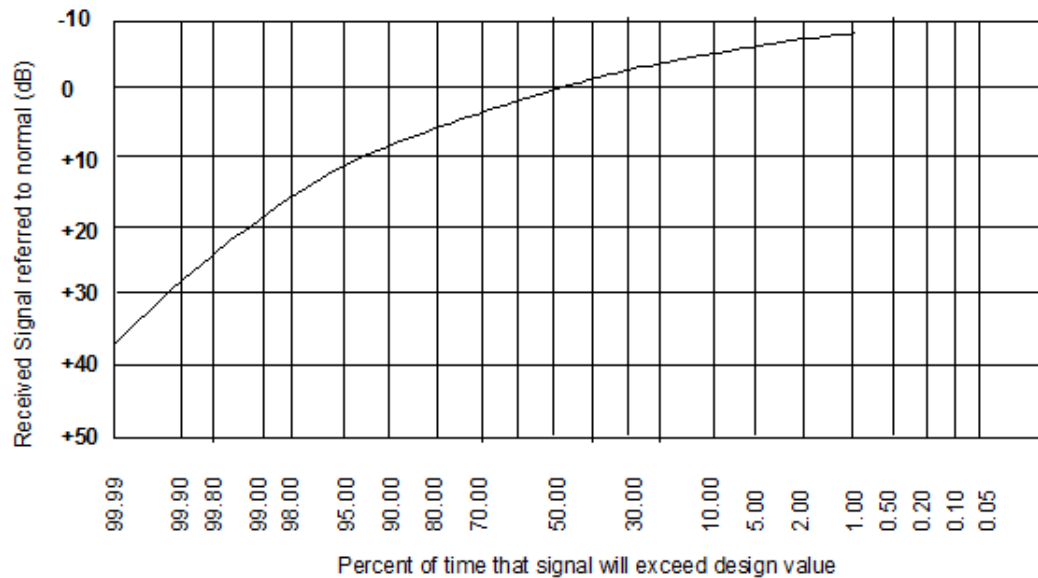
- **Antenna heights**
- **Terrain environment (urban or rural, etc)**
- **Weather**

- 3) As a RF engineer of one of the cellular operator in Singapore, you are in charge of the setting up of some base station in Singapore Orchard Road. Consider the following given parameters:

- Frequency of operation = 900 MHz
- Base station transmitter power output P_T = unknown
- Base Station feeder loss, L_F = 2 dB
- Mobile station sensitivity, P_R = -113 dBm
- Base station transmitting antenna gain, G_T = 10 dB
- Mobile station receiving antenna gain, G_R = 3 dB
- Height of Base station, h_T = 40 m
- Height of Mobile station, h_R = 2 m

Assume that there are no other losses except for the path loss and fading. The Base station must provide reliable communication within a cell radius $d = 2$ km with a reliability of 99 %. Using the Okumura-Hata Model, calculate the following:

- a) The required fade margin for 99% reliability [Answer = 18 dB]
- b) Write down the system equation in terms of the given parameters
- c) The path loss, L [Answer = 134 dB]
- d) The required Base station transmitter power P_T [Answer = ~~26 dB~~ 28dbm]



(a) For the required fade margin of 99% reliability, an extra 18 dB is required.

Note: Fade Margin is like buying insurance for your luggage when you go traveling. The more you treasure your belongings, the more you demanded that the luggage should reach your destination intact and therefore the more you should pay for the insurance premium.

(b) $P_T - L_f + G_T - L_p - FM + G_R - L_f = P_R$

(c) L_p for the Okumura Hata model in an urban setting is given by

$$L_p = 69.55 + 26.16 \log f \text{ (MHz)} - 13.82 \log h_{te} - A(h_{re}) + [44.9 - 6.55 \log h_{te}] [\log d \text{ (km)}]$$

$$A(h_{re}) = 3.2 [\log(11.75 h_{re})]^2 - 4.97 \text{ (dB)}$$

$$= 3.2 [\log(11.75 \times 2)]^2 - 4.97 \text{ dB} = 1.045 \text{ dB}$$

$$L_p = 69.55 + 26.16 \log 900 - 13.82 \log 40 - 1.045 + [44.9 - 6.55 \log 40] [\log 2]$$

$$= 134 \text{ dB}$$

(d) $P_T - 2 + 10 - 134 - 18 + 3 - 0 = -113$

$$P_T = 2 - 10 + 134 + 18 - 3 - 113 = 28 \text{ dBm}$$

- 4) A mobile subscriber that is living at a distance, $d = 3.5 \text{ km}$ from a base station complained to the cellular operator that his mobile phone reception is not very good. You are asked to investigate to find out the problems and make the necessary recommendations. Details are as follows:

Frequency of operation, f	: 935.2 MHz
Base Station transmitted output power, P_T	: 20 W
Base Station transmitter antenna gain, G_T	: 10 dB
Base Station transmitter antenna height, h_{te}	: 50 metre
Mobile Station receiver sensitivity	: -100 dBm

Mobile Station receiver antenna gain, G_R : 3 dB

Mobile Station receiver antenna height, h_{re} : 1.6 metre

Assume that all other losses are negligible except for the path loss and Rayleigh fading.

Using the Okumura-Hata propagation model, determine the following:

- (i) The system equation in terms of the given parameters.

$$P_T + G_T - L_P - FM + G_R = P_R$$

- (ii) The path loss, L_P (dB).

$$\begin{aligned} A(h_{re}) &= 3.2 [\log(11.75h_{re})]^2 - 4.97 \text{ (dB)} \\ &= 3.2 [\log(11.75 \times 1.6)]^2 - 4.97 \\ &= 0.2251 \end{aligned}$$

$$\begin{aligned} L_P &= 69.55 + 26.16 \log f(\text{MHz}) - 13.82 \log h_{te} - A(h_{re}) + [44.9 - 6.55 \log h_{te}] [\log d \text{ (km)}] \\ &= 69.55 + 26.16 \log(935.2) - 13.82 \log(50) - 0.2251 + [44.9 - 6.55 \log 50] \log 3.5 = 141.94 \text{ dB} \end{aligned}$$

- (iii) The received signal, P_R (dBm) at the mobile receiver, assuming the base station must provide communication with 99 % reliability.

For 99% reliability, $FM = 18$ dB

$$P_T + G_T - L_P - FM + G_R = P_R$$

$$P_R = 10 \log(20/0.001) + 10 - 141.94 - 18 + 3 = -103.93 \text{ dBm}$$

- (iv) Based on your result from (iii), is the signal strength received at subscriber's mobile phone sufficient? Explain your answer.

Insufficient as the signal strength is **weaker than the minimum of -100 dBm** required.

Okumura Hata Propagation model:

$$\begin{aligned} L_P &= 69.55 + 26.16 \log f(\text{MHz}) - 13.82 \log h_{te} - A(h_{re}) \\ &\quad + [44.9 - 6.55 \log h_{te}] [\log d \text{ (km)}] \\ A(h_{re}) &= 3.2 [\log(11.75h_{re})]^2 - 4.97 \text{ (dB)} \end{aligned}$$